CLIMATE SEAL

Product carbon intelligence for supply chains

Supply Chain Carbon Management White Paper

A practical operating model for supplier PCF, ISO 14067 workflows, evidence chains, multi-scenario reuse, and low-cost scale-up.

ENGLISH VERSION | MAGAZINE-STYLE EDITION

Translated, edited, and redesigned from the supplied Chinese draft.

SUPPLY CHAIN CARBON MANAGEMENT WHITE PAPER

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EDITORIAL PROPOSITION

Make supplier carbon data usable: minimum fields in, auditable PCF out, reusable everywhere.

EXECUTIVE SUMMARY

Why this white paper matters

For brands with material Scope 3 emissions, supplier carbon data has moved from a reputational project into a hard management requirement. Without supplier-level product carbon footprints (PCF), brands struggle to identify reduction hotspots, satisfy customer and tender requirements, and defend disclosures under audit.

- SBTi alignment makes supplier data operational: when Scope 3 is material, near-term targets must cover a large share of the value chain. Supplier PCF becomes the mechanism for coverage, not a side report.
- The largest emissions often sit upstream. Brands that cannot see supplier materials, energy, and logistics data cannot prioritize procurement levers or low-cost reduction opportunities.
- Buyer pull works. Walmart, Apple, Unilever, and public procurement programs show that supplier engagement, clean energy, and PCF/EPD readiness can be scaled when the operating model is explicit.
- The bottleneck is supplier capability. SMEs face limited people, limited technical knowledge, fragmented portals, repeated questionnaires, and data-quality concerns.
- The practical answer is a supplier workflow around ISO 14067 PCF: no-training data intake, automated draft calculation, evidence-chain capture, audit-ready outputs, and multi-scenario reuse.

Coverage Rate	≥80%	auditable PCF
Draft Speed	≤8 h	from complete data
Data Quality	≥4/5	evidence-ready
Reuse Rate	≥70%	2+ scenarios
Audit Pass	≥95%	sample review
Unit Cost	~1%	vs. consultant baseline

The real pain is not reporting; it is supplier operating capacity.

1.1 The five pain points brands feel today

Pain point	What it looks like on the ground	Business cost
Supplier capability gap	Suppliers copy old spreadsheets, use spend-based estimates, or do not understand boundaries, functional units, factor sources, and uncertainty.	Repeated data requests, rejected submissions, delayed audits, and rising consultant cost.
Portal and questionnaire fatigue	The same supplier receives multiple portals, templates, accounts, and inconsistent instructions from different brands.	Lower response rates, incomplete data, poor comparability, and more manual clean-up for the brand.
Data-quality and restatement risk	Scope 3 numbers are revised because supplier data gaps and methodology differences accumulate at group level.	Audit queries, investor-communication pressure, disclosure restatement, and possible tender/customer risk.
Software-only solutions fall short	A portal can assign tasks, but it does not teach suppliers how to calculate, evidence, or explain PCF data.	Brands pay twice: software plus consultants, while suppliers lose trust after repeated rework.
Missed reduction upside	Without reliable supplier data, brands cannot locate no-regret levers in energy, packaging, logistics, or materials.	Lost cost savings, weaker procurement strategy, and missed green-finance or customer advantages.

1.2 From vicious cycle to virtuous cycle

Vicious cycle	Virtuous cycle
Multiple portals and templates → suppliers cannot calculate or fill → low-quality data → rework and consultants → high cost and long cycle → supplier resistance → coverage never scales.	Unified scope and minimum fields → no-training intake from BOM, energy, and logistics → ISO 14067-aligned modeling → evidence chain and factor traceability → PDF plus machine-readable export → brand dashboard for coverage, quality, hotspots, and reduction potential.

1.3 Benchmark signals

- Walmart Project Gigaton demonstrates the power of structured buyer-supplier action around energy, packaging, logistics, and other executable projects.
- Apple supplier clean energy shows how supplier-level clean electricity programs can become measurable, audited climate progress.
- Unilever links supplier climate programs with PCF disclosure, SBTi-aligned targets, and procurement dialogue.

The common pattern is simple: organize suppliers, define measurable actions, collect consistent evidence, and review progress through a management system.

Use PCF and ISO 14067 as the common language for supplier data.

2.1 Minimum viable PCF pack

The goal is to allow a large supplier base to deliver auditable, reusable PCF data without heavy training. ISO 14067 provides the product-level carbon-footprint language: goal and scope, functional unit, boundary, quantification, reporting, and links to the ISO 14040/44 life-cycle assessment framework.

Deliverable	Purpose
PCF report brief	Goal, scope, functional unit, system boundary, version, and key assumptions.
Three data tables	Materials/BOM, energy, and logistics with units and measurement rules.
Factor traceability table	Database/source, version, geography, time period, and data-quality notes.
Uncertainty and assumption register	Substitutions, approximations, sensitivity, and expected impact.
Version and change log	Boundary, parameter, factor, and calculation updates.
Outputs	Audit-friendly PDF plus machine-readable JSON/CSV for reuse and system integration.

2.2 The three minimum supplier tables

Table	Minimum fields	Why it matters
Materials / BOM	Material name or ID, quantity per functional unit, unit, composition if available, recycled content, supplier/site, production year, evidence notes.	Establishes the reference flow and enables factor matching by material, geography, and time.
Energy	Energy type, consumption, unit, grid region, purchased green power if applicable, data period, meter or invoice source.	Electricity geography and year are critical for factor choice and audit evidence.
Logistics	Origin, destination, distance, transport mode, load class, optional load factor, mass or quantity basis.	Transport mode and distance drive logistics emissions and scenario comparison.

2.3 Seven-step workflow

- 1. Define functional unit and system boundary.
- 2. Collect the three minimum tables through Excel, CSV, ERP export, or a simple form.
- 3. Parse and normalize the BOM: hierarchy, duplicate items, units, and reference flow.
- 4. Match emission factors by material, geography, and year; document substitutes and confidence.
- 5. Calculate PCF and stage-level contributions for hotspot analysis.
- 6. Build credibility through evidence, uncertainty, assumptions, representativeness, and sensitivity.
- 7. Export PDF and machine-readable files, with EPD and battery passport mapping when needed.

2.4 Quality control and return rules

- Header checks: legal units, functional-unit consistency, and data years within the accepted window.
- Factor traceability: every material, energy, and logistics record must include source, version, geography, and year.
- Assumptions: substitutes or approximations must be documented, with sensitivity notes when impact is material.
- Versioning: every parameter or factor change receives a version number, change note, timestamp, and export checksum.

2.5 Mapping PCF to EPD and battery passport

A robust PCF source pack reduces the work required for Environmental Product Declarations (EPD), tenders, customer templates, and regulated product transparency. EPDs require PCR alignment, modular life-cycle reporting, verification, and program-operator metadata. Battery passports require factory, batch, material-origin, stage contribution, and public disclosure elements. The same source data can support both when modeled with traceability from the beginning.

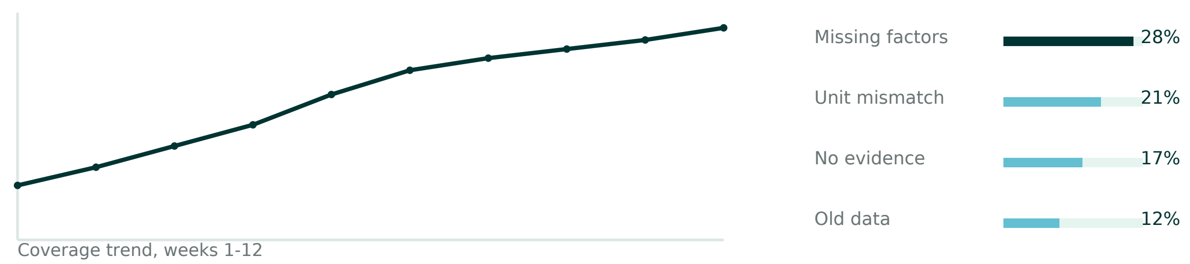
Make the system measurable: coverage, quality, reuse, audit, time, and cost.

3.1 North-star target and KPI system

Within three months, a brand should be able to run a repeatable pilot without increasing supplier burden. Within twelve months, the operating target is: auditable PCF coverage of 80% or more, stable data quality, and multi-scenario reuse across brand audit, tender, EPD, battery passport, and financing.

KPI	Definition	First-year target
Coverage Rate (CR)	Suppliers with an auditable PCF pack ÷ target suppliers.	≥80%
Time to Draft / Approval	Complete data to draft; draft to buyer approval.	T2D ≤8h; T2A ≤5 working days
Unit Cost (UC)	Compliance cost ÷ approved SKUs.	Single-digit RMB/SKU as marginal cost
Data Quality Score (DQS)	Evidence, factor traceability, representativeness, scope consistency, assumptions.	≥4/5
Reuse Rate (RR)	PCF packs successfully used in at least two scenarios.	≥70%
Audit Pass Rate (APR)	First-pass rate in random sample or third-party review.	≥95%

Pilot dashboard prototype



3.2 Dashboard design

- Executive view: CR, RR, APR, T2D/T2A, median unit cost, weekly trend, and top failure reasons.
- Operations view: supplier completion, return reasons, action logs, BOM completeness, factor hit rate, and task board by SLA.
- Quality/audit view: DQS radar, sampling records, corrective actions, second-pass rate, version changes, and recalculation rate.

3.3 Governance

- Freeze the functional unit, system boundary, units, precision, and template version before data collection.

- Apply strict return rules for missing fields, missing factor traceability, old data, undocumented assumptions, and unit mismatch.
- Sample 10-20% of suppliers weekly, with higher sampling for complex categories or low DQS suppliers.
- Freeze versions and checksums before EPD, battery passport, tender, or customer-template reuse.

Convert benchmark lessons into repeatable brand actions.

4.1 What the benchmarks teach

Benchmark	Signal	Action for brands
Walmart Project Gigaton	Large-scale supplier mobilization around energy, packaging, logistics, and other executable projects.	Put reduction action lists into supplier collaboration and review them annually.
Apple supplier clean energy	Supplier renewable electricity can be tracked as a measurable transition program.	Combine grid power, PPAs/RECs, and on-site renewables; keep contracts, invoices, and generation evidence.
Unilever supplier climate program	PCF becomes part of supplier climate expectations and procurement dialogue.	Score suppliers by PCF availability, data quality, and reuse readiness.
Volvo EX90 battery passport	Product transparency is moving toward batch, material, and carbon-footprint disclosure.	Capture factory/batch-specific PCF and material-origin evidence early.
Government procurement and EPD	EPDs increasingly support market access and low-carbon material thresholds.	Prioritize PCF-to-EPD mapping for construction, materials, and public-procurement categories.

4.2 Worked example A: product PCF skeleton

Functional unit: one aluminum profile component, 0.385 kg. Boundary: cradle-to-gate, A1-A3, including upstream material, manufacturing energy, internal transport, and one optional outbound logistics segment.

- A1 material emissions = aluminum mass × aluminum emission factor.
- A3 manufacturing energy = electricity consumption × local grid factor, with green power listed separately if claimed.
- A2 / distribution = tonne-kilometers × transport-mode factor, where applicable.
- Total PCF = A1 + A3 (+ A2), accompanied by evidence, factor source/version, geography, time period, substitutions, and uncertainty.

4.3 Worked example B: logistics scenario

For the same 1 kg product shipped from East China to Europe, the workflow calculates tonne-kilometers and applies mode-specific factors. Air freight can be orders of magnitude higher than ocean freight, so the result supports both tender explanation and logistics optimization. The final number must use the correct route, load assumptions, and current factor tables.

4.4 ROI logic

Baseline consultant model	Automated minimum workflow
2-3.5 person-days/SKU for collection, modeling, and rework; about RMB 500/SKU in the illustrative model; 500 SKUs can take 10+ weeks.	Three-table intake, automated checks, BOM parsing, factor matching, and exception review; illustrative marginal cost around RMB 8/SKU and draft generation in hours.

The absolute cost depends on industry complexity, but the direction is consistent: standardize input, automate routine work, reserve human review for exceptions, and make each PCF pack reusable.

No-training in, trusted assessment, multi-format reuse out.

5.1 Pain point to solution element

Brand pain	Solution element	Business metric
Suppliers cannot calculate, fill, or staff the task.	No-training collection through three templates, examples, unit checks, and lightweight forms.	Response rate ↑; time to draft ↓
Multiple portals create fatigue.	One model, multi-format export to PDF, machine-readable files, EPD, battery passport, tender, and customer templates.	Reuse rate ↑; repeated requests ↓
Consultant dependence and high unit cost.	Automated modeling, factor matching, and return rules; human review only for exceptions.	Unit cost approaches ~1% of baseline
Data not trusted or audit-ready.	Evidence chains, factor traceability, version logs, uncertainty and sensitivity notes.	DQS ≥4/5; APR ≥95%
Cannot scale.	Operations dashboard and KPI governance across coverage, quality, reuse, audit, time, and cost.	CR ≥80% in steady state

5.2 Architecture

- Data intake: Excel, CSV, API, ERP/MES/energy/logistics exports, and mobile-compatible forms mapped into the three-table semantics.
- Semantic standardization: unit normalization, material hierarchy cleanup, duplicate merging, reference-flow construction, and suspicious-value flags.
- Factor matching engine: product/site-specific factors first, then regional industry databases, then global generic factors with confidence and substitute rules.
- ISO 14067 calculation core: fixed functional unit and boundary, staged outputs, hotspot contribution, repeatable logs, and parameter snapshots.
- Trusted assessment: evidence files, assumptions, uncertainty, change logs, checksums, and one-click audit package.
- Reuse export: PDF report, JSON/CSV, EPD/PCR mappings, battery passport fields, and customer/tender templates.

5.3 How the 1% cost logic works

Cost falls when the fixed system cost is spread across many SKUs and the exception rate decreases. The formula is: unit cost = system amortization per approved SKU + exception labor × exception rate per approved SKU. As supplier waves scale from 100 to 1,000 and rules improve, the system cost dilutes and the exception rate narrows.

5.4 Procurement and organizational linkage

- Require a PCF minimum viable pack within a defined SLA in RFPs and framework agreements.
- Set quality gates such as DQS ≥4/5, first-pass audit requirements, and corrective-action rules.

- Define reuse rights so the same data pack can support tender, EPD, battery passport, customer audit, or financing with confidentiality controls.
- Incentivize suppliers through preferred status, payment-term optimization, joint case publication, or support resources.

Align decision owners, budget logic, objections, and roadmaps.

6.1 Who approves and who executes

The decision triangle is ESG/sustainability, procurement, and finance. Execution is shared by operations/data and IT. Third-party verifiers or consultants should confirm methods and sampling results, not replace supplier-first data production.

6.2 Three board-ready business cases

Method	Argument
Unit-cost case	Compare automated PCF workflow with consultant/person-day baseline.
Risk-cost case	Quantify reduced audit queries, restatements, rework, and disclosure inconsistency.
Market-access case	Estimate the effect of EPD, customer audit, tender readiness, or green-finance advantages on order value and win probability.

6.3 Common objections and responses

Objection	Response
Can suppliers just fill a platform?	Assigning a task is not the same as enabling calculation. The workflow provides minimum templates, auto-checks, factor traceability, and audit packages.
Self-reported data is not credible.	Every record is tied to evidence, factor source/version, representativeness, assumptions, and sampling review.
Will this expose confidential data?	Collect only minimum necessary fields; use ranges or redaction where appropriate; keep role-based access and read-only audit interfaces.
Will procurement be overloaded?	Procurement mobilizes and escalates; technical judgment sits with operations, ESG methodology, and quality/audit.
Will consultants be displaced?	No. Their value moves to method review, sampling verification, and complex technical decisions.

6.4 30/60/90/180/365-day roadmap

Time	Milestone
30 days	Launch templates and return rules; invite 50-100 key suppliers; produce 10-20 SKU drafts and first approvals.
60 days	Run sampling and corrections; complete EPD/battery passport field mapping; achieve two reuse scenarios.
90 days	Pilot acceptance: CR ≥60%, DQS ≥4/5, APR ≥90%, RR ≥60%, T2D ≤8h, T2A ≤5 days.
180 days	Scale to 300-500 suppliers; release category factor packs; connect one or two green-finance cases.
365 days	Steady state: CR ≥80%, RR ≥70%, APR ≥95%; PCF enters procurement scoring, financing, and tenders as a hard metric.

6.5 Communication copy

Customer/tender line: We have established a supplier PCF mechanism that supports one-time modeling, multi-format export, audit readiness, and reuse for EPD, battery passport, and customer templates.

Supplier message: Three tables, no heavy training, automated modeling, and one data pack reused across multiple brand needs. Suppliers that meet the quality gate enter the preferred pool and may be featured in joint reduction cases.

The detailed operating text a team can run with.

7.0 Brand and document rules

- Primary color: Climate Seal deep green #003432; supporting neutrals and light mint; charts default to restrained single-color treatment unless emphasis is needed.
- Header: Climate Seal tree-and-water logo at top left; footer: version, date, document owner, and page number.
- File naming: {Org}_{DocType}_{SKU-or-Topic}_{vX.Y}_{YYYYMMDD}.{ext}.

7.1 RACI

Role	Responsibility
ESG Methodology (A/R)	Owns scope, functional unit, boundary, scoring rules, and verifier alignment.
Procurement (A/R)	Mobilizes suppliers, manages clauses and incentives, escalates exceptions.
Data Ops (R)	Collects data, parses BOM, matches factors, calculates, exports, and logs versions.
Quality/Internal Audit (C/R)	Maintains return rules, sampling plans, corrective actions, and audit package standards.
IT/Security (C)	Manages permissions, logs, data residency, integrations, checksum, and signatures.
Legal (C)	Reviews terms, reuse authorization, privacy, confidentiality, and IP boundaries.
Finance (C)	Tracks budget, unit cost, ROI, green-finance opportunities, and supplier incentives.

7.2 Data model and validation

Table	Required fields	Automatic checks
Materials	Item ID, material name, functional-unit quantity, unit, production year. Recommended: grade, supplier site, region, recycled content.	Unit validity, quantity >0, data year within 3 years, functional-unit consistency.
Energy	Energy type, consumption, unit, location country, optional grid region.	kWh for power, correct fuel units, legal time window, mandatory geography.
Logistics	Mode, distance, distance unit, product quantity or mass basis, from/to country.	km/mi and kg/t/pcs checks, distance >0, mandatory origin and destination.

7.3 Factor matching rules

- Priority order: product/site-specific factors, then regional industry factors, then global generic factors.
- Each matched factor records source, version, geography, time representativeness, unit, and data-quality note.
- Substitutions must state why they are used, confidence level, and expected direction of impact.
- High-impact substitutions trigger sensitivity analysis and quality review.

7.4 Audit package checklist

- PCF summary report with functional unit, system boundary, assumptions, and version.
- Three data tables, factor traceability table, assumption register, uncertainty notes, and sensitivity results.
- Evidence folder: invoices, meter records, production reports, logistics records, contracts, database extracts, or screenshots.
- Change log, calculation parameters, export checksums, and reviewer sign-off.

7.5 Acceptance and supplier email templates

Invitation: To support customer, compliance, and reduction-planning needs, we are collecting minimum fields through three templates. The system will generate an audit-ready draft PCF pack. Please submit by [date]; questions can be sent by reply or checked in the FAQ.

Return for correction: Your submission did not pass the mandatory checks: [reason]. Please update the attached template and resubmit. Common reasons include data year older than three years, missing factor source/version/geography/year, or invalid units.

Acceptance and reuse: Your PCF pack has passed the initial review, version [vX.Y], checksum [ID]. It may be reused for EPD, customer audit, tender, or other approved scenarios. Method changes will be communicated through the scope-alignment checklist.

CLOSING STATEMENT

The end goal is not another report. It is a repeatable supplier operating system for low-burden, low-cost, auditable product carbon data.